

Inverting a Real-Argument Fibonacci Function

Log-Periodic Transseries, Exact Multivariate Coefficients, and a General Product-Reversion Calculus

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Abstract

We study the nonnegative even interpolation

$$\mathcal{F}(x) = \sqrt{\frac{2}{5}} \sqrt{\cosh(2x \operatorname{arccsch} 2) - \cos(\pi x)}, \quad x \in \mathbb{R},$$

which satisfies $\mathcal{F}(n) = F_n$ for every nonnegative integer n . The map is not globally one-to-one, but it is strictly increasing on $[2, \infty)$. Hence it has a positive outer inverse $X_+(y)$ for $y \geq 1$, and evenness gives the negative outer inverse $X_-(y) = -X_+(y)$. We determine their complete large- y transseries.

Put $\lambda = 2 \log \varphi$, $\rho = \pi/\lambda$, $R = 5y^2$, and $u = e^{-\lambda x}$. The inverse equation admits the factorization

$$Ru = (1 - u^{1-i\rho})(1 - u^{1+i\rho}).$$

This identity converts the inverse problem into an analytic bivariate reversion at the ordinary origin. Multivariate Lagrange–Good inversion then gives the explicit all-orders formula

$$X_+(y) = \frac{\log R}{\lambda} + \sum_{\substack{j,k \geq 0 \\ j+k \geq 1}} \frac{(-1)^{j+k+1}}{\lambda s_{jk}} \binom{s_{jk}}{j} \binom{s_{jk}}{k} R^{-s_{jk}}, \quad s_{jk} = j + k - i\rho(j - k).$$

The series is not merely formal: it converges for all sufficiently large y . Pairing conjugate terms reorganizes it as

$$X_+(y) = \xi + \sum_{n \geq 1} D_n(\theta) R^{-n}, \quad \xi = \frac{\log R}{\lambda}, \quad \theta = \pi\xi,$$

where each D_n is an explicit real trigonometric polynomial whose harmonics have the same parity as n . We derive the first sectors, a triangular recurrence, uniform remainder bounds, differentiation rules, a hypergeometric–elliptic formula for the angular mean, and the exact Banach-space radius of uniform convergence. Numerical tests include the integer checkpoints $X_+(2) = 3$ and $X_+(5) = 5$.

The second part develops a reusable theory for inverses of finite products

$$R = e^{\lambda x} \prod_{r=1}^m (1 - c_r e^{-\lambda \alpha_r x})^{\beta_r}, \quad \Re \alpha_r > 0,$$

obtaining a closed multi-index coefficient formula. We then extend the method to arbitrary analytic perturbations of a dominant exponential and to mixed carriers such as $e^{\lambda x} x^\mu$, where Lambert W and polynomial–logarithmic coefficient fields enter. The resulting framework explains systematically why conjugate complex exponents generate log-periodic or quasiperiodic inverse transseries.

Main result

Let

$$\lambda = 2 \log \varphi, \quad \rho = \frac{\pi}{\lambda}, \quad R = 5y^2,$$

and define

$$\xi = \frac{\log R}{\lambda} = \log_{\varphi}(\sqrt{5}y), \quad \theta = \rho \log R = \pi \xi, \quad q = R^{-1}.$$

For the positive outer branch, as $y \rightarrow +\infty$,

$$\begin{aligned} X_+(y) = & \xi + \frac{2q}{\lambda} \cos \theta \\ & - \frac{q^2}{\lambda} (2 + \cos 2\theta + 2\rho \sin 2\theta) \\ & + \frac{q^3}{\lambda} \left[(6 - \rho^2) \cos \theta + 5\rho \sin \theta + \left(\frac{2}{3} - 3\rho^2 \right) \cos 3\theta + 3\rho \sin 3\theta \right] + \mathcal{O}(q^4). \end{aligned}$$

Equivalently, the leading correction is

$$X_+(y) = \log_{\varphi}(\sqrt{5}y) + \frac{\cos(\pi \log_{\varphi}(\sqrt{5}y))}{5(\log \varphi)y^2} + \mathcal{O}(y^{-4}).$$

The complete coefficient formula is

$$X_+(y) = \frac{\log R}{\lambda} + \sum_{\substack{j,k \geq 0 \\ j+k \geq 1}} \frac{(-1)^{j+k+1}}{\lambda s_{jk}} \binom{s_{jk}}{j} \binom{s_{jk}}{k} R^{-s_{jk}}, \quad s_{jk} = j + k - i\rho(j - k).$$

For every fixed N , truncation at $j + k = N$ has error $\mathcal{O}(R^{-N-1}) = \mathcal{O}(y^{-2N-2})$. The negative outer branch is $X_-(y) = -X_+(y)$.

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1 Introduction

The ordinary Fibonacci sequence is discrete, while inversion is naturally a continuous or complex-analytic operation. There are many inequivalent ways to continue Fibonacci numbers away from the integers. The continuation in this article is distinguished by three elementary features: it is real and nonnegative on the real axis, it is even, and it agrees exactly with the nonnegative Fibonacci numbers at nonnegative integers. These features are purchased at the cost of global injectivity and, at the origin, ordinary analyticity. Neither issue obstructs inversion at large values.

The asymptotic problem has an unusual scale. The dominant exponential gives $x \sim \log_\varphi(\sqrt{5}y)$, but the first correction is not a power of $1/\log y$. It has size y^{-2} and carries a periodic function of $\log y$. Thus it is beyond all inverse-logarithmic orders while still algebraic in y^{-1} . In the language of transseries, the relevant monomials are complex powers

$$R^{-n+i\rho h} = R^{-n} e^{i h \rho \log R}, \quad n \in \mathbb{N},$$

organized into finite Fourier packets at every real order R^{-n} . Conjugation makes the final answer real.

A direct Taylor reversion of the original square-root formula is possible, but it hides the decisive structure. Squaring and introducing $u = e^{-\lambda x}$ reveals an exact factorization into two conjugate factors. The inverse then becomes a bivariate fixed-point problem with an ordinary analytic germ at the origin. This makes the multivariate Lagrange–Good theorem almost perfectly adapted to the problem and produces a closed coefficient formula.

The article has two goals. First, it gives a complete treatment of the specific Fibonacci interpolation: branches, explicit sectors, convergence, error certification, computation, and numerical checks. Second, it isolates the mechanism and proves a multi-factor inversion theorem applicable to a large class of exponential products. Arbitrary analytic perturbations and mixed dominant carriers are treated afterward.

2 The real-argument Fibonacci modulus

2.1 Constants and equivalent forms

Let

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad a = \operatorname{arccsch} 2, \quad \lambda = 2a, \quad \omega = \pi, \quad \rho = \frac{\omega}{\lambda}. \quad (2.1)$$

Because $\sinh a = 1/2$, the positive number $t = e^a$ satisfies $t - t^{-1} = 1$, hence $t = \varphi$. Therefore

$$a = \log \varphi, \quad \lambda = 2 \log \varphi. \quad (2.2)$$

Numerically,

$$\lambda = 0.9624236501192068949955 \dots, \quad \rho = 3.2642513026364969 \dots$$

Define

$$\mathcal{F}(x) = \sqrt{\frac{2}{5}} \sqrt{\cosh(\lambda x) - \cos(\omega x)}, \quad x \in \mathbb{R}, \quad (2.3)$$

where the nonnegative square root is understood. Squaring and expanding the hyperbolic cosine gives

$$\mathcal{F}(x)^2 = \frac{1}{5} \left(e^{\lambda x} + e^{-\lambda x} - 2 \cos(\omega x) \right). \quad (2.4)$$

Since $\lambda = 2 \log \varphi$, this can also be written as the modulus of a complex Binet interpolation:

$$\mathcal{F}(x) = \frac{1}{\sqrt{5}} \left| \varphi^x - e^{i\pi x} \varphi^{-x} \right|. \quad (2.5)$$

The modulus in Equation (2.5) is essential: the complex expression inside it is an analytic continuation, whereas $\mathcal{F}$ itself is an even, nonnegative real function.

Proposition 2.1 (Interpolation of Fibonacci numbers). *For every integer $n \geq 0$,*

$$\mathcal{F}(n) = F_n,$$

where $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$.

Proof. Binet's formula is

$$F_n = \frac{\varphi^n - (-\varphi^{-1})^n}{\sqrt{5}}.$$

It is nonnegative for $n \geq 0$, and therefore

$$\begin{aligned} F_n^2 &= \frac{1}{5} (\varphi^{2n} + \varphi^{-2n} - 2(-1)^n) \\ &= \frac{1}{5} (e^{\lambda n} + e^{-\lambda n} - 2 \cos(\pi n)) = \mathcal{F}(n)^2. \end{aligned}$$

Taking nonnegative square roots proves the claim. $\square$

Remark 2.2. The extension is not the usual signed extension to negative indices. It is even, so $\mathcal{F}(-n) = F_n$, whereas the discrete signed convention gives $F_{-n} = (-1)^{n+1} F_n$. This distinction matters when the two outer inverse branches are named.

2.2 Basic geometry and the outer inverse branches

Equation (2.3) immediately shows that $\mathcal{F}$ is even and that its only real zero is $x = 0$. Near the origin,

$$\mathcal{F}(x) = \sqrt{\frac{\lambda^2 + \omega^2}{5}} |x| + \mathcal{O}(|x|^3), \quad (2.6)$$

so the even interpolation has a cusp there.

For $x > 0$, differentiation of Equation (2.4) gives

$$\mathcal{F}'(x) = \frac{\lambda \sinh(\lambda x) + \omega \sin(\omega x)}{5\mathcal{F}(x)}. \quad (2.7)$$

The oscillatory sine can create a small nonmonotone region, but exponential growth eventually dominates it.

Proposition 2.3 (Monotonicity of the outer branch). *Let*

$$x_* = \frac{1}{\lambda} \operatorname{arsinh}\left(\frac{\omega}{\lambda}\right) = 1.972992734136994 \dots \quad (2.8)$$

Then $\mathcal{F}'(x) > 0$ for every $x > x_*$. In particular, $\mathcal{F}$ is strictly increasing on $[2, \infty)$, and it maps that interval bijectively onto $[1, \infty)$.

Proof. By Equation (2.7), the sign is the sign of its numerator. Since $\sin(\omega x) \geq -1$,

$$\lambda \sinh(\lambda x) + \omega \sin(\omega x) \geq \lambda \sinh(\lambda x) - \omega.$$

The right-hand side is positive exactly when $x > x_*$. Moreover, $x_* < 2$ and $\mathcal{F}(2) = F_2 = 1$. Since $\mathcal{F}(x) \rightarrow \infty$, the stated bijection follows. $\square$

Definition 2.4 (Outer inverse branches). For $y \geq 1$, let $X_+(y) \in [2, \infty)$ be the unique solution of $\mathcal{F}(x) = y$. Define

$$X_-(y) = -X_+(y).$$

These are the positive and negative outer inverse branches.

On the entire real line, $\mathcal{F}$ is not globally one-to-one. Numerically, the two nontrivial critical points in $(1, 2)$ are approximately

$$1.132379068\dots, \quad 1.719250886\dots,$$

with function values $1.013859885\dots$ and $0.911172684\dots$, respectively. Consequently, some moderate values have additional inner preimages. This finite-scale geometry disappears from the large-value problem: because $[0, 2]$ is compact, every sufficiently large y has exactly the two real preimages $X_+(y)$ and $X_-(y)$.

A point requiring care

The phrase “the inverse of $\mathcal{F}$ ” is ambiguous unless a branch is specified. Every asymptotic formula below first refers to X_+ ; the negative real branch follows by evenness. Complex branches require an additional logarithm index and are discussed in Section 14.

3 Exact normalization and factorization

3.1 The squared inverse equation

Fix the positive outer branch and put

$$R = 5y^2. \tag{3.1}$$

The equation $\mathcal{F}(x) = y$ becomes

$$R = e^{\lambda x} + e^{-\lambda x} - 2 \cos(\omega x). \tag{3.2}$$

For positive real x , introduce

$$u = e^{-\lambda x} \in (0, 1), \quad \rho = \frac{\omega}{\lambda}. \tag{3.3}$$

Using the real logarithm of u ,

$$e^{i\omega x} = u^{-i\rho}, \quad e^{-i\omega x} = u^{i\rho}.$$

Therefore Equation (3.2) is equivalent to

$$R = u^{-1} + u - u^{i\rho} - u^{-i\rho}. \tag{3.4}$$

Multiplication by u produces the key identity.

Theorem 3.1 (Conjugate factorization). *With*

$$\alpha = 1 - i\rho, \quad \bar{\alpha} = 1 + i\rho, \tag{3.5}$$

the positive inverse equation is exactly

$$Ru = (1 - u^\alpha)(1 - u^{\bar{\alpha}}). \tag{3.6}$$

Proof. Expanding the right-hand side gives

$$(1 - u^{1-i\rho})(1 - u^{1+i\rho}) = 1 - u^{1-i\rho} - u^{1+i\rho} + u^2,$$

which is precisely u times the right-hand side of Equation (3.4). $\square$

This factorization is the structural reason the inverse coefficients admit a closed formula. Without it one still obtains a recursion, but the two oscillatory pieces appear unrelated. In factorized form they are a conjugate pair.

3.2 Baseline, phase, and the normalized implicit equation

Define

$$q = R^{-1}, \quad \xi = \frac{\log R}{\lambda}, \quad \theta = \rho \log R = \omega \xi. \quad (3.7)$$

The dominant balance in Equation (3.2) is $R \sim e^{\lambda x}$, hence $x \sim \xi$. Write

$$x = \xi + \delta. \quad (3.8)$$

Since $e^{\lambda \xi} = R$, division of Equation (3.2) by R yields the exact scalar equation

$$H(\delta; q, \theta) := e^{\lambda \delta} + q^2 e^{-\lambda \delta} - 2q \cos(\theta + \omega \delta) - 1 = 0. \quad (3.9)$$

At $q = 0$, this reduces to $e^{\lambda \delta} = 1$, and the real germ is $\delta = 0$. Furthermore,

$$\frac{\partial H}{\partial \delta}(0; 0, \theta) = \lambda \neq 0.$$

The analytic implicit-function theorem therefore gives a unique small solution for $|q|$ sufficiently small, uniformly in real θ . Equation (3.9) already proves the existence of an expansion in powers of q whose coefficients are trigonometric polynomials. The factorization will do much more: it will calculate every coefficient in one line.

3.3 Bivariate lift

From Equation (3.6), set

$$V = \frac{u}{q}, \quad A = u^\alpha, \quad B = u^{\bar{\alpha}}, \quad X = q^\alpha, \quad Y = q^{\bar{\alpha}}. \quad (3.10)$$

Then

$$V = (1 - A)(1 - B), \quad (3.11)$$

and, because $u = qV$,

$$A = XV^\alpha, \quad B = YV^{\bar{\alpha}}. \quad (3.12)$$

Finally,

$$\delta = -\frac{1}{\lambda} \log V. \quad (3.13)$$

Equations (3.11) to (3.13) are an ordinary bivariate analytic reversion problem at $(X, Y) = (0, 0)$. For positive real q ,

$$X = qe^{i\theta}, \quad Y = qe^{-i\theta}. \quad (3.14)$$

Thus every monomial $X^j Y^k$ becomes $q^{j+k} e^{i(j-k)\theta}$.

Structural checkpoint

The large variable y has disappeared from the analytic reversion itself. It enters only through the small radial variable $q = (5y^2)^{-1}$ and the logarithmic angle $\theta = \rho \log(5y^2)$. The inverse correction is an analytic bivariate germ restricted to the spiral $(X, Y) = (qe^{i\theta}, qe^{-i\theta})$.

4 The log-periodic transseries algebra

4.1 Complex powers and Fourier packets

For the present problem the natural monomials are

$$R^{-s_{jk}}, \quad s_{jk} = \alpha j + \bar{\alpha} k = j + k - i\rho(j - k), \quad j, k \in \mathbb{N}_0. \quad (4.1)$$

Their absolute sizes are $R^{-(j+k)}$. If

$$n = j + k, \quad h = j - k, \quad (4.2)$$

then $|h| \leq n$, $h \equiv n \pmod{2}$, and

$$R^{-s_{jk}} = q^n e^{ih\theta}. \quad (4.3)$$

At each fixed radial order q^n , only finitely many Fourier harmonics are possible.

Definition 4.1 (Fibonacci log-periodic algebra). Let $\mathcal{A}_\rho$ be the set of formal series

$$T(q, \theta) = \sum_{n \geq 0} D_n(\theta) q^n, \quad (4.4)$$

where

$$D_n(\theta) = \sum_{\substack{|h| \leq n \\ h \equiv n \pmod{2}}} d_{n,h} e^{ih\theta} \quad (4.5)$$

and, for real series, $d_{n,-h} = \overline{d_{n,h}}$. The valuation of a nonzero series is the least n for which $D_n \neq 0$.

The cone and parity conditions in Equation (4.5) are closed under multiplication. The resulting q -adic metric makes $\mathcal{A}_\rho$ complete. If T has positive valuation, then analytic substitutions such as $\exp T$, $\log(1 + T)$, and $\cos(\theta + T)$ are well defined by their ordinary Taylor series: only finitely many terms contribute to any prescribed radial order. Consequently Equation (3.9) has a unique formal solution $\delta \in q\mathcal{A}_\rho$, independently of convergence questions.

4.2 Differentiation

The algebra is also stable under the Euler derivative. Since $q = R^{-1}$ and $\theta = \rho \log R$,

$$R \frac{d}{dR} (q^n e^{ih\theta}) = (-n + i\rho h) q^n e^{ih\theta} = -s_{jk} R^{-s_{jk}}. \quad (4.6)$$

In the original variable y , because $R = 5y^2$,

$$y \frac{d}{dy} (q^n e^{ih\theta}) = 2(-n + i\rho h) q^n e^{ih\theta}. \quad (4.7)$$

Thus differentiation does not change radial valuation. Ordinary derivatives add the expected powers of y^{-1} .

4.3 Why this is a transseries rather than an ordinary power series

If q and θ were independent, Equation (4.4) would simply be a power series with trigonometric-polynomial coefficients. Along the physical curve, however, $\theta = -\rho \log q$, so every Fourier factor is a complex power of q . The scale is therefore a finitely generated grid of complex powers. It is well based by the real valuation $\Re s_{jk} = j + k$, even though the complex exponents are not linearly ordered by their imaginary parts.

Relative to the coarse expansion scale generated by $\log y$, $\log \log y$, and inverse logarithms, every nonconstant sector of $\mathcal{A}_\rho$ is beyond all orders:

$$y^{-2} = o((\log y)^{-M}) \quad \text{for every fixed } M > 0.$$

This explains why a purely polynomial–logarithmic analysis recovers only the carrier ξ and misses the oscillatory corrections.

Section summary

The correct coefficient ring is not merely $\mathbb{R}[[y^{-2}]]$. It is a complete valued algebra of Fourier packets in $\log y$. The bivariate variables X and Y linearize its support: $X^j Y^k$ corresponds to the complex-power monomial $R^{-s_{jk}}$.

5 Multivariate Lagrange–Good inversion

5.1 The coefficient theorem

We use the following form of multivariate Lagrange inversion. For a multi-index $\mathbf{n} = (n_1, \dots, n_m)$, write $\mathbf{Q}^{\mathbf{n}} = \prod_r Q_r^{n_r}$ and similarly for $\mathbf{A}^{\mathbf{n}}$.

Theorem 5.1 (Lagrange–Good coefficient formula). *Suppose analytic germs $A_r = A_r(\mathbf{Q})$ are determined near the origin by*

$$A_r = Q_r \Phi_r(\mathbf{A}), \quad r = 1, \dots, m, \quad \Phi_r(\mathbf{0}) \neq 0. \quad (5.1)$$

Then for every analytic germ f ,

$$[\mathbf{Q}^{\mathbf{n}}]f(\mathbf{A}(\mathbf{Q})) = [\mathbf{A}^{\mathbf{n}}]f(\mathbf{A}) \prod_{r=1}^m \Phi_r(\mathbf{A})^{n_r} \Delta(\mathbf{A}), \quad (5.2)$$

where

$$\Delta(\mathbf{A}) = \det \left(\delta_{rs} - A_s \frac{\partial}{\partial A_s} \log \Phi_r(\mathbf{A}) \right)_{r,s=1}^m. \quad (5.3)$$

The theorem can be proved by a multivariate residue change of variables. The determinant is the Jacobian correction for $Q_r = A_r / \Phi_r(\mathbf{A})$. We will not merely invoke the formula: in the present system its determinant has rank one, and the entire coefficient calculation collapses to generalized binomial coefficients.

5.2 Evaluation for the Fibonacci system

For Equation (3.12),

$$\Phi_1 = V^\alpha, \quad \Phi_2 = V^{\bar{\alpha}}, \quad V = (1 - A)(1 - B).$$

Because

$$\frac{\partial \log V}{\partial A} = -\frac{1}{1-A}, \quad \frac{\partial \log V}{\partial B} = -\frac{1}{1-B},$$

the Good matrix is

$$\begin{pmatrix} 1 + \frac{\alpha A}{1-A} & \frac{\alpha B}{1-B} \\ \frac{\bar{\alpha} A}{1-A} & 1 + \frac{\bar{\alpha} B}{1-B} \end{pmatrix}.$$

It is an identity matrix plus a rank-one matrix, so

$$\Delta(A, B) = 1 + \frac{\alpha A}{1-A} + \frac{\bar{\alpha} B}{1-B}. \quad (5.4)$$

For $j, k \geq 0$, define

$$s_{jk} = \alpha j + \bar{\alpha} k. \quad (5.5)$$

Introduce the auxiliary coefficient

$$P_{jk}(t) = [A^j B^k] V^t \Delta(A, B). \quad (5.6)$$

The next lemma is the computational core.

Lemma 5.2 (Binomial collapse). *For $(j, k) \neq (0, 0)$,*

$$P_{jk}(t) = (-1)^{j+k} \binom{t}{j} \binom{t}{k} \left(1 - \frac{s_{jk}}{t}\right), \quad (5.7)$$

where

$$\binom{z}{n} = \frac{z(z-1) \cdots (z-n+1)}{n!}, \quad \binom{z}{0} = 1. \quad (5.8)$$

The apparent singularity at $t = 0$ in Equation (5.7) is removable when the expression is interpreted as the polynomial obtained from Equation (5.6).

Proof. The contribution of the constant term 1 in Equation (5.4) is

$$[A^j B^k] (1-A)^t (1-B)^t = (-1)^{j+k} \binom{t}{j} \binom{t}{k}.$$

For the A -term in the determinant, the coefficient is

$$\begin{aligned} & \alpha [A^{j-1}] (1-A)^{t-1} [B^k] (1-B)^t \\ &= \alpha (-1)^{j+k-1} \binom{t-1}{j-1} \binom{t}{k} = -(-1)^{j+k} \binom{t}{j} \binom{t}{k} \frac{\alpha j}{t}. \end{aligned}$$

The B -term similarly contributes the factor $-\bar{\alpha} k/t$. Adding the three pieces gives Equation (5.7). $\square$

5.3 Differentiating at the forced zero

Apply Theorem 5.1 to the auxiliary function $f(A, B) = V^\tau$. Since $\Phi_1^j \Phi_2^k = V^{s_{jk}}$,

$$[X^j Y^k] V(A(X, Y), B(X, Y))^\tau = P_{jk}(\tau + s_{jk}). \quad (5.9)$$

Differentiating at $\tau = 0$ gives

$$[X^j Y^k] \log V = P'_{jk}(s_{jk}). \quad (5.10)$$

The factor $1 - s_{jk}/t$ in Equation (5.7) vanishes at $t = s_{jk}$, so derivatives of the binomial factors disappear and

$$P'_{jk}(s_{jk}) = \frac{(-1)^{j+k}}{s_{jk}} \binom{s_{jk}}{j} \binom{s_{jk}}{k}. \quad (5.11)$$

Combining this with Equation (3.13) proves the explicit coefficient formula.

Theorem 5.3 (Exact bivariate inverse germ). *In a neighborhood of $(X, Y) = (0, 0)$, the inverse correction is the convergent bivariate series*

$$\delta(X, Y) = \sum_{\substack{j, k \geq 0 \\ j+k \geq 1}} C_{jk} X^j Y^k, \quad (5.12)$$

where

$$C_{jk} = \frac{(-1)^{j+k+1}}{\lambda s_{jk}} \binom{s_{jk}}{j} \binom{s_{jk}}{k}, \quad s_{jk} = j + k - i\rho(j - k). \quad (5.13)$$

Proof. The analytic implicit-function theorem applied to Equation (3.12) gives analytic germs $A(X, Y)$ and $B(X, Y)$, so V and δ are analytic. Equations (3.13), (5.10) and (5.11) calculate every nonconstant Taylor coefficient. The constant term vanishes because $V(0, 0) = 1$. $\square$

Remark 5.4 (Why the denominator is harmless). For $(j, k) \neq (0, 0)$, $\Re s_{jk} = j + k > 0$, hence $s_{jk} \neq 0$. No resonance or small-denominator problem occurs in this positive cone. The imaginary part records phase; it cannot cancel the positive real valuation.

6 Complete inverse transseries

6.1 Complex-power form

Substitute Equation (3.14) into Equation (5.12). Since $X^j Y^k = R^{-s_{jk}}$, we obtain the central result.

Theorem 6.1 (All-orders inverse of the Fibonacci modulus). *Let X_+ be the outer inverse from Theorem 2.4. For all sufficiently large positive y ,*

$$X_+(y) = \frac{\log R}{\lambda} + \sum_{\substack{j, k \geq 0 \\ j+k \geq 1}} \frac{(-1)^{j+k+1}}{\lambda s_{jk}} \binom{s_{jk}}{j} \binom{s_{jk}}{k} R^{-s_{jk}} \quad (6.1)$$

with

$$R = 5y^2, \quad \lambda = 2 \log \varphi, \quad \rho = \frac{\pi}{\lambda}, \quad s_{jk} = j + k - i\rho(j - k). \quad (6.2)$$

The series converges to the exact branch. Independently of convergence, it is a complete Poincaré asymptotic expansion: for every fixed N ,

$$X_+(y) = \frac{\log R}{\lambda} + \sum_{1 \leq j+k \leq N} C_{jk} R^{-s_{jk}} + \mathcal{O}(R^{-N-1}). \quad (6.3)$$

The negative branch satisfies

$$X_-(y) = -X_+(y). \quad (6.4)$$

Proof. The bivariate series in Theorem 5.3 converges in a polydisc about the origin. Along the physical curve $(X, Y) = (qe^{i\theta}, qe^{-i\theta})$, both variables tend to zero as $y \rightarrow \infty$, so the series converges and solves the exact normalized equation. Uniqueness of the small real solution identifies it with the outer branch. Truncation by total degree leaves an analytic remainder of total degree at least $N + 1$; because $|X| = |Y| = q = R^{-1}$, this is $\mathcal{O}(R^{-N-1})$. Evenness proves Equation (6.4). $\square$

6.2 Real Fourier-sector form

The coefficients satisfy

$$C_{kj} = \overline{C_{jk}}. \quad (6.5)$$

Consequently terms with (j, k) and (k, j) pair to a real expression. For $n \geq 1$ and admissible h , define

$$\widehat{D}_{n,h} = \frac{(-1)^{n+1}}{\lambda(n - i\rho h)} \binom{n - i\rho h}{(n+h)/2} \binom{n - i\rho h}{(n-h)/2}. \quad (6.6)$$

Then put

$$D_n(\theta) = \sum_{\substack{|h| \leq n \\ h \equiv n \pmod{2}}} \widehat{D}_{n,h} e^{ih\theta}. \quad (6.7)$$

Corollary 6.2 (Real log-periodic sectors). *The positive outer inverse can be written*

$$X_+(y) = \xi + \sum_{n \geq 1} D_n(\theta) q^n, \quad q = (5y^2)^{-1}, \quad \xi = \log_\varphi(\sqrt{5}y), \quad \theta = \pi\xi. \quad (6.8)$$

Each D_n is real for real θ , has Fourier degree at most n , and contains only harmonics congruent to n modulo 2. Moreover,

$$D_n(\theta + \pi) = (-1)^n D_n(\theta). \quad (6.9)$$

Proof. Set $n = j + k$ and $h = j - k$ in Equation (6.1). Equation (6.5) implies $\widehat{D}_{n,-h} = \overline{\widehat{D}_{n,h}}$, hence reality. The support conditions follow from $j = (n+h)/2$ and $k = (n-h)/2$. Finally, $e^{ih\pi} = (-1)^h = (-1)^n$ for every admissible h , proving Equation (6.9). $\square$

6.3 The first three sectors

Direct simplification of Equation (6.7) gives

$$D_1(\theta) = \frac{2}{\lambda} \cos \theta, \quad (6.10)$$

$$D_2(\theta) = -\frac{1}{\lambda} (2 + \cos 2\theta + 2\rho \sin 2\theta), \quad (6.11)$$

$$D_3(\theta) = \frac{1}{\lambda} \left[(6 - \rho^2) \cos \theta + 5\rho \sin \theta + \left(\frac{2}{3} - 3\rho^2 \right) \cos 3\theta + 3\rho \sin 3\theta \right]. \quad (6.12)$$

Thus

$$\begin{aligned} X_+(y) &= \xi + \frac{2q}{\lambda} \cos \theta - \frac{q^2}{\lambda} (2 + \cos 2\theta + 2\rho \sin 2\theta) \\ &\quad + \frac{q^3}{\lambda} \left[(6 - \rho^2) \cos \theta + 5\rho \sin \theta + \left(\frac{2}{3} - 3\rho^2 \right) \cos 3\theta + 3\rho \sin 3\theta \right] + \mathcal{O}(q^4). \end{aligned} \quad (6.13)$$

The leading form in the original variable is

$$X_+(y) = \log_\varphi(\sqrt{5}y) + \frac{\cos(\pi \log_\varphi(\sqrt{5}y))}{5(\log \varphi)y^2} + \mathcal{O}(y^{-4}). \quad (6.14)$$

The fourth sector is compactly represented by positive-frequency coefficients. If

$$D_n(\theta) = d_{n,0} + 2\Re \sum_{\substack{1 \leq h \leq n \\ h \equiv n \pmod{2}}} d_{n,h} e^{ih\theta}, \quad (6.15)$$

with the constant term omitted for odd n , then

$$d_{4,0} = -\frac{9}{\lambda}, \quad (6.16)$$

$$d_{4,2} = -\frac{2(\lambda - i\omega)(2\lambda - i\omega)(3\lambda - 2i\omega)}{3\lambda^4}, \quad (6.17)$$

$$d_{4,4} = -\frac{(\lambda - 4i\omega)(\lambda - 2i\omega)(3\lambda - 4i\omega)}{12\lambda^4}. \quad (6.18)$$

The general formula Equation (6.6) is usually preferable to fully expanded trigonometric expressions at higher orders.

Remark 6.3 (Amplitude and phase form). Every conjugate pair can equivalently be written

$$2|\widehat{D}_{n,h}|q^n \cos(h\theta + \arg \widehat{D}_{n,h}).$$

The phase shifts are therefore algebraic consequences of the complex binomial coefficients; they are not independent fitting parameters.

Section summary

The outer inverse consists of the logarithmic carrier ξ plus a convergent hierarchy of log-periodic sectors. The sector of radial order $q^n = (5y^2)^{-n}$ contains precisely the harmonics $n, n-2, \dots, -n$. Formula Equation (6.6) gives every Fourier coefficient directly.

7 Recursive construction and rigorous remainders

The closed formula is ideal for theory, but a triangular recurrence is often more convenient for symbolic implementation. Return to the normalized scalar equation Equation (3.9) and seek

$$\delta(q, \theta) = \sum_{n \geq 1} D_n(\theta) q^n. \quad (7.1)$$

Suppose $D_1, \dots, D_{n-1}$ are known and set

$$\delta_{<n}(q, \theta) = \sum_{r=1}^{n-1} D_r(\theta) q^r. \quad (7.2)$$

Proposition 7.1 (Triangular sector recurrence). *For every $n \geq 1$,*

$$D_n(\theta) = -\frac{1}{\lambda} [q^n] \left(e^{\lambda \delta_{<n}} + q^2 e^{-\lambda \delta_{<n}} - 2q \cos(\theta + \omega \delta_{<n}) - 1 \right). \quad (7.3)$$

Here $[q^n]$ means extraction of the coefficient of q^n , treating θ as an independent parameter.

Proof. Substitute $\delta = \delta_{<n} + D_n q^n + \mathcal{O}(q^{n+1})$ into Equation (3.9). In $e^{\lambda \delta}$, the unknown sector contributes $\lambda D_n q^n$. The cosine term is multiplied by q , so its dependence on $D_n q^n$ begins at order q^{n+1} ; the second exponential is multiplied by q^2 , so its dependence begins still later. Thus the coefficient equation at order q^n is linear with coefficient λ , yielding Equation (7.3). $\square$

7.1 Bell-polynomial form

The recurrence can be expanded using complete exponential Bell polynomials. For example, if $E(q) = \exp(\lambda\delta(q)) = \sum_{n \geq 0} E_n q^n$, then

$$E_n = \frac{1}{n!} B_n(1!\lambda D_1, 2!\lambda D_2, \dots, n!\lambda D_n), \quad (7.4)$$

where B_n is the complete Bell polynomial. The Fourier cone is preserved under all products appearing in these Bell polynomials. A similar expansion of $\exp(i\omega\delta)$ handles the cosine through

$$\cos(\theta + \omega\delta) = \Re\left(e^{i\theta} e^{i\omega\delta}\right).$$

This gives a purely algebraic, finite computation at each order.

The closed coefficient formula and the triangular recurrence are useful for different purposes. Formula [Equation \(5.13\)](#) exposes arithmetic and asymptotic structure; recurrence [Equation \(7.3\)](#) is easy to truncate, simplify, and evaluate numerically without manipulating generalized binomial functions of large complex arguments.

7.2 Residual certification

For $N \geq 0$, let

$$\delta_N(q, \theta) = \sum_{n=1}^N D_n(\theta) q^n, \quad \delta_0 = 0. \quad (7.5)$$

By construction,

$$H(\delta_N; q, \theta) = \mathcal{O}(q^{N+1}). \quad (7.6)$$

The derivative of H is

$$H_\delta(\delta; q, \theta) = \lambda e^{\lambda\delta} - \lambda q^2 e^{-\lambda\delta} + 2\omega q \sin(\theta + \omega\delta). \quad (7.7)$$

For $|q|$ and $|\delta|$ small, this is bounded away from zero uniformly in real θ .

Theorem 7.2 (Uniform remainder estimate). *For each $N \geq 0$, there exist $q_N > 0$ and $C_N > 0$ such that, for real θ and $0 < q < q_N$,*

$$|\delta(q, \theta) - \delta_N(q, \theta)| \leq C_N q^{N+1}. \quad (7.8)$$

Equivalently,

$$X_+(y) - \left(\xi + \sum_{n=1}^N D_n(\theta) (5y^2)^{-n} \right) = \mathcal{O}(y^{-2N-2}). \quad (7.9)$$

For every fixed ordinary derivative order $m \geq 0$,

$$\frac{d^m}{dy^m} \left[X_+(y) - \xi - \sum_{n=1}^N D_n(\theta) (5y^2)^{-n} \right] = \mathcal{O}(y^{-2N-2-m}). \quad (7.10)$$

Proof. Equation [Equation \(7.6\)](#) and the uniform lower bound for $|H_\delta|$ imply [Equation \(7.8\)](#) by the mean-value theorem on the real branch, or by the analytic inverse-function theorem with a uniform compact parameter $\theta \in \mathbb{R}/2\pi\mathbb{Z}$. Since $q = (5y^2)^{-1}$, this gives [Equation \(7.9\)](#). The exact solution is analytic in (X, Y) near the origin, so the normalized remainder is analytic in q with smooth periodic dependence on θ . Applying

$$y \frac{d}{dy} = -2q \frac{\partial}{\partial q} + 2\rho \frac{\partial}{\partial \theta}$$

preserves radial order. Converting Euler derivatives to ordinary derivatives adds one factor y^{-1} per differentiation, proving [Equation \(7.10\)](#). $\square$

7.3 Newton acceleration in the transseries algebra

A second computational route is Newton iteration applied to [Equation \(3.9\)](#):

$$\delta^{\text{new}} = \delta - \frac{H(\delta; q, \theta)}{H_\delta(\delta; q, \theta)}. \quad (7.11)$$

All operations are performed in the truncated algebra $\mathcal{A}_\rho/(q^{N+1})$. If the current approximation has residual valuation r , the next Newton iterate ordinarily has residual valuation at least $2r$. Starting from $\delta = 0$, repeated doubling therefore computes N sectors in $\mathcal{O}(\log N)$ Newton stages, assuming fast truncated multiplication. The triangular recurrence remains simpler for hand calculation, while Newton iteration is attractive for very high order.

Algorithm 7.3 (Certified high-order inverse evaluation). For a prescribed y , sector order N , and target precision:

1. compute $R = 5y^2$, $q = R^{-1}$, $\xi = \log R/\lambda$, and $\theta = \pi\xi$;
2. generate $D_1, \dots, D_N$ either from [Equation \(6.6\)](#) or [Equation \(7.3\)](#);
3. evaluate $x_N = \xi + \sum_{n=1}^N D_n(\theta)q^n$;
4. compute the exact residual $r_N = \mathcal{F}(x_N) - y$, or preferably the squared residual in [Equation \(3.2\)](#);
5. convert the residual to an a posteriori error estimate using a lower bound for $\mathcal{F}'$ near x_N , or perform one full-precision Newton correction.

8 Coefficient structure

8.1 Diagonal, edge, and mean sectors

Formula [Equation \(6.6\)](#) distinguishes several natural families. The edge harmonics $h = \pm n$ correspond to $(j, k) = (n, 0)$ and $(0, n)$. The central harmonic $h = 0$ exists only for even $n = 2m$ and corresponds to $j = k = m$. Since $s_{m,m} = 2m$, its coefficient is particularly simple:

$$\widehat{D}_{2m,0} = -\frac{1}{2m\lambda} \binom{2m}{m}^2. \quad (8.1)$$

The first values are

$$-\frac{2}{\lambda}, \quad -\frac{9}{\lambda}, \quad -\frac{200}{3\lambda}, \quad -\frac{1225}{2\lambda}, \quad \dots$$

These constants are the angular means of the even sectors.

8.2 Hypergeometric and elliptic resummation of the mean

For a periodic function, define

$$\langle f \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta. \quad (8.2)$$

Averaging [Equation \(6.8\)](#) and using [Equation \(8.1\)](#) gives

$$\langle \delta \rangle(q) = -\frac{1}{\lambda} \sum_{m \geq 1} \frac{1}{2m} \binom{2m}{m}^2 q^{2m}. \quad (8.3)$$

This subseries has the closed hypergeometric form

$$\langle \delta \rangle (q) = -\frac{2q^2}{\lambda} {}_4F_3 \left(\begin{matrix} 1, 1, \frac{3}{2}, \frac{3}{2} \\ 2, 2, 2 \end{matrix}; 16q^2 \right). \quad (8.4)$$

Let $\mathbf{K}(k)$ denote the complete elliptic integral in modulus notation,

$$\mathbf{K}(k) = \int_0^{\pi/2} \frac{dt}{\sqrt{1 - k^2 \sin^2 t}}. \quad (8.5)$$

The classical generating function

$$\sum_{m \geq 0} \binom{2m}{m}^2 q^{2m} = \frac{2}{\pi} \mathbf{K}(4q) \quad (8.6)$$

then implies

$$q \frac{d}{dq} \langle \delta \rangle (q) = -\frac{1}{\lambda} \left(\frac{2}{\pi} \mathbf{K}(4q) - 1 \right). \quad (8.7)$$

The mean sector has radius $|q| < 1/4$. The complete transseries has a smaller uniform radius because off-diagonal harmonics grow more rapidly.

Remark 8.1. Averaging over the logarithmic phase is not the same operation as evaluating at a typical large y : the physical angle $\theta = \rho \log R$ keeps rotating. Nevertheless, the mean isolates a distinguished exactly summable subsector and reveals a direct elliptic-integral connection.

8.3 Large-order growth and the exact uniform radius

Set $j = np$, $k = n(1 - p)$ heuristically and define

$$\sigma(p) = 1 - i\rho(2p - 1), \quad 0 \leq p \leq 1. \quad (8.8)$$

With principal logarithms and the convention $0 \log 0 = 0$, put

$$\begin{aligned} g_\rho(p) = & \Re[2\sigma \operatorname{Log} \sigma - (\sigma - p) \operatorname{Log}(\sigma - p) - (\sigma - 1 + p) \operatorname{Log}(\sigma - 1 + p)] \\ & - p \log p - (1 - p) \log(1 - p). \end{aligned} \quad (8.9)$$

The function is continuous and symmetric under $p \mapsto 1 - p$.

Theorem 8.2 (Coefficient growth and uniform convergence radius). *Let*

$$G_\rho = \exp \left(\max_{0 \leq p \leq 1} g_\rho(p) \right), \quad q_* = G_\rho^{-1}. \quad (8.10)$$

Then

$$\limsup_{n \rightarrow \infty} \max_{0 \leq j \leq n} |C_{j, n-j}|^{1/n} = G_\rho, \quad (8.11)$$

and, with the supremum norm on continuous 2π -periodic functions,

$$\limsup_{n \rightarrow \infty} \|D_n\|_\infty^{1/n} = G_\rho. \quad (8.12)$$

Consequently the Banach-valued power series $\sum_{n \geq 1} D_n(\theta) q^n$ has exact radius q_ for uniform convergence in θ .*

Proof. Write the generalized binomial coefficients in Equation (5.13) as Gamma quotients and apply Stirling's formula with $j/n \rightarrow p$. The terms proportional to $n \log n$ and n cancel, leaving precisely $g_\rho(p)$ as the exponential rate. Uniform Stirling estimates on closed subintervals, followed by continuous endpoint limits, give Equation (8.11).

Every $C_{j,n-j}$ is a Fourier coefficient of D_n , hence

$$\max_j |C_{j,n-j}| \leq \|D_n\|_\infty.$$

Conversely,

$$\|D_n\|_\infty \leq \sum_{j=0}^n |C_{j,n-j}| \leq (n+1) \max_j |C_{j,n-j}|.$$

Taking n -th roots proves Equation (8.12); the Cauchy–Hadamard formula gives the radius. $\square$

For the Fibonacci value $\rho = \pi/(2 \log \varphi)$, numerical maximization gives two symmetric maximizers,

$$p_* = 0.0934868615594596 \dots, \quad 1 - p_* = 0.9065131384405404 \dots, \quad (8.13)$$

and

$$G_\rho = 10.1243628948610529 \dots, \quad (8.14)$$

$$q_* = 0.09877164720237184 \dots \quad (8.15)$$

Therefore the Fourier-sector expansion converges absolutely and uniformly whenever

$$q = (5y^2)^{-1} < q_*; \quad \text{in particular, whenever} \quad y > 1.422980175185941 \dots \quad (8.16)$$

This sufficient-and-sharp threshold concerns uniform convergence as a power series with values in $C(\mathbb{R}/2\pi\mathbb{Z})$. Special phases may have larger pointwise domains; no such enlargement is needed for the large-argument theory.

The dominant large-order coefficients are not the angular means. The mean has per-radial-order growth 4^n , while the full norm grows like G_ρ^n with $G_\rho \approx 10.1244$. The maximizing normalized frequency is

$$\frac{|h|}{n} = |2p_* - 1| = 0.813026276881081 \dots,$$

so high-order growth is driven by off-center harmonics.

Section summary

The inverse transseries is a genuinely convergent expansion, not merely a formal asymptotic device. Its exact uniform radial radius is determined by a one-dimensional entropy-like maximization derived from complex Stirling asymptotics. For every integer Fibonacci checkpoint beginning with $y = F_3 = 2$, the physical value $q = (5y^2)^{-1}$ lies comfortably inside the uniform disk.

9 Numerical verification

The explicit series can be tested without any ambiguity by substituting its partial sums into the original function. Define

$$X_{+,N}(y) = \xi + \sum_{n=1}^N D_n(\theta) q^n, \quad X_{+,0}(y) = \xi. \quad (9.1)$$

The following values were computed with high-precision arithmetic. The entries E_N are absolute errors $|X_{+,N}(y) - X_+(y)|$. At $y = 2$ and $y = 5$, the exact inverse values are integers because $2 = F_3$ and $5 = F_5$.

Table 1: Exact inverse values and absolute errors of sector truncations.

y	exact $X_+(y)$
2	3
5	5
10	6.45779309028821164945...
100	11.24218977013280389416...

y	E_0	E_1	E_2	E_3
2	1.1269603×10^{-1}	1.5236037×10^{-2}	2.9618737×10^{-3}	2.2607545×10^{-3}
5	1.6827815×10^{-2}	2.2634536×10^{-4}	1.8596489×10^{-5}	9.6297844×10^{-7}
10	5.4518532×10^{-4}	1.1352239×10^{-5}	1.5425214×10^{-7}	2.1471593×10^{-9}
100	3.0101615×10^{-5}	3.5619587×10^{-9}	$2.9715886 \times 10^{-13}$	$5.7169682 \times 10^{-18}$

y	E_4	E_6	E_8
2	3.7691613×10^{-4}	1.2868198×10^{-4}	1.5141006×10^{-5}
5	2.9886773×10^{-8}	$1.8233102 \times 10^{-11}$	$6.5804309 \times 10^{-13}$
10	$3.0170906 \times 10^{-11}$	$6.0001812 \times 10^{-15}$	$1.1659689 \times 10^{-18}$
100	$4.3013834 \times 10^{-21}$	$1.4868585 \times 10^{-28}$	$2.1225057 \times 10^{-36}$

The convergence is phase dependent, so errors need not decrease monotonically at every adjacent order. For example, at $y = 2$ the order-10 and order-11 errors are approximately 1.1640×10^{-6} and 1.4004×10^{-6} , respectively. The all-orders estimate concerns the magnitude of the remainder after a fixed truncation as $y \rightarrow \infty$, while convergence for a fixed admissible y follows from the power-series theorem. By $y = 10$, each additional sector already contributes roughly two more decimal digits; at $y = 100$, the leading carrier is accurate to five decimal places and a few sectors give extremely high precision.

9.1 Integer checkpoints

For every integer $n \geq 2$, [Theorem 2.1](#) and [Theorem 2.3](#) imply

$$X_+(F_n) = n. \quad (9.2)$$

For $n \geq 3$, one has $q = (5F_n^2)^{-1} \leq 1/20 < q_*$, so the uniform convergence theorem applies. Substituting $y = F_n$ into [Equation \(6.1\)](#) therefore yields the convergent identity:

$$n = \log_{\varphi}(\sqrt{5} F_n) + \sum_{\substack{j,k \geq 0 \\ j+k \geq 1}} C_{jk} (5F_n^2)^{-s_{jk}}. \quad (9.3)$$

This is not a tautological restatement of Binet's formula at the level of individual sectors. It decomposes the exact difference between n and the logarithmic Binet estimate into a convergent hierarchy of log-periodic complex-power corrections.

At integer n , Binet's formula gives

$$\sqrt{5} F_n = \varphi^n (1 - (-1)^n \varphi^{-2n}),$$

so

$$\log_{\varphi}(\sqrt{5} F_n) = n + \frac{1}{\log \varphi} \log(1 - (-1)^n \varphi^{-2n}). \quad (9.4)$$

The leading sector in [Equation \(6.14\)](#) cancels the first term of the logarithm in [Equation \(9.4\)](#); all higher sectors complete the exact cancellation. This provides a stringent symbolic and numerical check on signs and phase conventions.

10 Wolfram Language implementation

The user-supplied interpolation is already expressed in Wolfram Language, so it is natural to give an implementation in the same notation. The following code evaluates the exact coefficient formula and groups it by radial order.

Listing 1: Closed coefficients and truncated inverse transseries.

```
ClearAll[lam, rho, s, c, dn, inverseSeries];

lam = 2 ArcCsch[2]; (* = 2 Log[GoldenRatio] *)
rho = Pi/lam;

s[j_Integer?NonNegative, k_Integer?NonNegative] :=
  (1 - I rho) j + (1 + I rho) k;

c[j_Integer?NonNegative, k_Integer?NonNegative] /; j + k > 0 :=
  (-1)^(j + k + 1)/(lam s[j, k]) *
  Binomial[s[j, k], j] Binomial[s[j, k], k];

dn[n_Integer?Positive, theta_] :=
  FullSimplify[
    ComplexExpand[
      Sum[c[j, n - j] Exp[I (2 j - n) theta], {j, 0, n}],
      TargetFunctions -> {Re, Im}
    ],
    Assumptions -> Element[theta, Reals]
  ];

inverseSeries[y_, nmax_Integer?NonNegative] := Module[
  {r = 5 y^2, q, xi, theta},
  q = 1/r;
  xi = Log[r]/lam;
  theta = Pi xi;
  xi + Sum[dn[n, theta] q^n, {n, 1, nmax}]
];
```

For numerical validation, one may compare against a high-precision root of the original function.

Listing 2: Direct outer inverse and residual check.

```
ClearAll[f, outerInverse, check];

f[x_] := Sqrt[2/5] Sqrt[Cosh[lam x] - Cos[Pi x]];

outerInverse[y_?NumericQ, wp_: 80] := Module[{x0},
  x0 = N[Log[5 y^2]/lam, wp];
  x /. FindRoot[
    f[x] == SetPrecision[y, wp],
    {x, x0},
    WorkingPrecision -> wp,
    AccuracyGoal -> Floor[wp/2],
```

```

PrecisionGoal -> Floor[wp/2]
]
];

check[y_?NumericQ, nmax_Integer?NonNegative, wp_: 80] := Module[
{xa, xt},
xa = N[inverseSeries[SetPrecision[y, wp], nmax], wp];
xt = outerInverse[y, wp];
<|
"Approximation" -> xa,
"DirectRoot" -> xt,
"AbsoluteError" -> Abs[xa - xt],
"FunctionResidual" -> Abs[f[xa] - y]
|>
];

```

The recurrence [Equation \(7.3\)](#) can also be implemented by repeatedly forming a truncated Series. The closed formula is less stateful and naturally parallelizes over pairs (j, k) ; the recurrence is usually faster after simplification when one needs all sectors up to a large common order.

10.1 Stable numerical evaluation of high-order coefficients

For moderate n , Wolfram Language's exact generalized Binomial expression is reliable. At large n , direct products can overflow or suffer cancellation in machine arithmetic. Three remedies are standard:

1. evaluate with arbitrary precision and use LogGamma to form Gamma quotients in logarithmic scale;
2. pair (j, k) with (k, j) before summing, so every contribution is explicitly real;
3. use the triangular recurrence or Newton iteration in the real Fourier basis, avoiding large intermediate complex binomials.

For a requested numerical value rather than symbolic coefficients, a few transseries sectors followed by one Newton step is typically the most stable and economical procedure.

Part I

General inversion theory

11 Finite exponential products

The Fibonacci factorization is a special case of a broad class whose inverses admit equally explicit coefficients. Let $m \geq 1$, let $\lambda \neq 0$, and consider a chosen sector in which $\Re(\lambda x) \rightarrow +\infty$. Fix parameters

$$c_r, \beta_r \in \mathbb{C}, \quad \alpha_r \in \mathbb{C}, \quad \Re \alpha_r > 0, \quad r = 1, \dots, m. \quad (11.1)$$

Branches of powers are chosen by analytic continuation from factors near 1.

Consider the normalized product map

$$R = e^{\lambda x} \prod_{r=1}^m \left(1 - c_r e^{-\lambda \alpha_r x}\right)^{\beta_r}. \quad (11.2)$$

A nonzero constant prefactor A causes no difficulty: replace R by R/A .

11.1 Fixed-point normalization

Set

$$u = e^{-\lambda x}, \quad q = R^{-1}. \quad (11.3)$$

Then Equation (11.2) is

$$Ru = \prod_{r=1}^m (1 - c_r u^{\alpha_r})^{\beta_r}. \quad (11.4)$$

Introduce

$$A_r = c_r u^{\alpha_r}, \quad Q_r = c_r q^{\alpha_r}, \quad V = \frac{u}{q} = \prod_{r=1}^m (1 - A_r)^{\beta_r}. \quad (11.5)$$

Then

$$A_r = Q_r V^{\alpha_r}, \quad r = 1, \dots, m, \quad (11.6)$$

and the inverse correction is

$$x = \frac{\log R}{\lambda} - \frac{1}{\lambda} \log V. \quad (11.7)$$

This is the same architecture as the Fibonacci problem, with an arbitrary number of factors and arbitrary exponents.

For a multi-index $\mathbf{n} = (n_1, \dots, n_m) \in \mathbb{N}_0^m$, write

$$|\mathbf{n}| = \sum_{r=1}^m n_r, \quad S_{\mathbf{n}} = \sum_{r=1}^m \alpha_r n_r, \quad \mathbf{c}^{\mathbf{n}} = \prod_{r=1}^m c_r^{n_r}. \quad (11.8)$$

Because every $\Re \alpha_r > 0$, one has $\Re S_{\mathbf{n}} > 0$ for every nonzero $\mathbf{n}$.

Theorem 11.1 (Closed inverse of a finite exponential product). *For $|R|$ sufficiently large in the selected sector, the inverse branch of Equation (11.2) satisfying $x \sim \lambda^{-1} \log R$ is*

$$x(R) = \frac{\log R}{\lambda} + \sum_{\mathbf{n} \in \mathbb{N}_0^m \setminus \{\mathbf{0}\}} \frac{(-1)^{|\mathbf{n}|+1}}{\lambda S_{\mathbf{n}}} \left[\prod_{r=1}^m \binom{\beta_r S_{\mathbf{n}}}{n_r} c_r^{n_r} \right] R^{-S_{\mathbf{n}}}. \quad (11.9)$$

The multi-series converges near $R = \infty$ in its natural variables $Q_r = c_r R^{-\alpha_r}$, and it is a complete asymptotic transseries when ordered by $\Re S_{\mathbf{n}}$.

Proof. Apply Theorem 5.1 to Equation (11.6), where

$$\Phi_r(\mathbf{A}) = V(\mathbf{A})^{\alpha_r}, \quad V(\mathbf{A}) = \prod_{s=1}^m (1 - A_s)^{\beta_s}.$$

The Good matrix is

$$M_{rs} = \delta_{rs} + \alpha_r \frac{\beta_s A_s}{1 - A_s}. \quad (11.10)$$

It is $I + \mathbf{a}\mathbf{b}^T$, where $a_r = \alpha_r$ and $b_s = \beta_s A_s / (1 - A_s)$. The matrix determinant lemma gives

$$\Delta(\mathbf{A}) = 1 + \sum_{r=1}^m \frac{\alpha_r \beta_r A_r}{1 - A_r}. \quad (11.11)$$

Define

$$P_{\mathbf{n}}(t) = [\mathbf{A}^{\mathbf{n}}] V(\mathbf{A})^t \Delta(\mathbf{A}). \quad (11.12)$$

The constant part of the determinant contributes

$$B_{\mathbf{n}}(t) = (-1)^{|\mathbf{n}|} \prod_{r=1}^m \binom{\beta_r t}{n_r}. \quad (11.13)$$

For the summand indexed by r in Equation (11.11), use

$$\beta_r \binom{\beta_r t - 1}{n_r - 1} = \frac{n_r}{t} \binom{\beta_r t}{n_r}.$$

Its sign is opposite to Equation (11.13), so summing all determinant terms gives

$$P_{\mathbf{n}}(t) = B_{\mathbf{n}}(t) \left(1 - \frac{S_{\mathbf{n}}}{t}\right). \quad (11.14)$$

For an auxiliary exponent τ , Good inversion gives

$$[Q^{\mathbf{n}}]V(A(Q))^{\tau} = P_{\mathbf{n}}(\tau + S_{\mathbf{n}}). \quad (11.15)$$

Differentiation at $\tau = 0$ yields the coefficient of $\log V$. The factor in parentheses in Equation (11.14) vanishes at $t = S_{\mathbf{n}}$, hence

$$[Q^{\mathbf{n}}]\log V = \frac{(-1)^{|\mathbf{n}|}}{S_{\mathbf{n}}} \prod_{r=1}^m \binom{\beta_r S_{\mathbf{n}}}{n_r}. \quad (11.16)$$

Now use Equation (11.7) and $Q^{\mathbf{n}} = c^{\mathbf{n}} R^{-S_{\mathbf{n}}}$ to obtain Equation (11.9). Analytic convergence near $Q = 0$ follows from the implicit-function theorem. $\square$

11.2 Checks and special cases

Example 11.2 (One factor). For

$$R = e^{\lambda x} (1 - ce^{-\lambda \alpha x})^{\beta},$$

Equation (11.9) reduces to

$$x = \frac{\log R}{\lambda} + \sum_{n \geq 1} \frac{(-1)^{n+1}}{\lambda \alpha n} \binom{\beta \alpha n}{n} c^n R^{-\alpha n}. \quad (11.17)$$

When $\alpha = \beta = 1$, the original equation is $R = e^{\lambda x} - c$, whose exact inverse is

$$x = \frac{1}{\lambda} \log(R + c) = \frac{\log R}{\lambda} + \frac{1}{\lambda} \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \left(\frac{c}{R}\right)^n.$$

Formula Equation (11.17) reproduces it because $\binom{n}{n} = 1$.

Example 11.3 (Fibonacci conjugate pair). Take $m = 2$,

$$c_1 = c_2 = 1, \quad \beta_1 = \beta_2 = 1, \quad \alpha_1 = 1 - i\rho, \quad \alpha_2 = 1 + i\rho.$$

Then $S_{(j,k)} = s_{jk}$, and Equation (11.9) becomes exactly Equation (6.1).

Example 11.4 (Purely real corrections). If all α_r, c_r, β_r are real and $\alpha_r > 0$, every exponent $S_{\mathbf{n}}$ is real. The result is an ordinary generalized power series in real powers of R^{-1} , with no logarithmic oscillation. Log-periodicity appears precisely when nonreal exponents occur.

11.3 Reality, periodicity, and quasiperiodicity

Suppose the parameter list is closed under complex conjugation, with conjugate values of c_r , α_r , and β_r . Multi-indices are then paired by conjugation, and the inverse transseries is real on the positive real axis. A monomial with $S_n = a - ib$ takes the form

$$R^{-S_n} = R^{-a} e^{ib \log R}. \quad (11.18)$$

If all frequencies b lie in an integer multiple of one basic frequency, the coefficients are periodic functions of $\log R$. If several basic frequencies are rationally independent, the coefficient functions are quasiperiodic on a finite-dimensional torus. The same inversion theorem handles both cases without modification; only the presentation of the monomial group changes.

11.4 Support and local finiteness

Let

$$\Gamma = \{S_n : \mathbf{n} \in \mathbb{N}_0^m\}. \quad (11.19)$$

Because $\min_r \Re \alpha_r > 0$, for every $A > 0$ there are only finitely many multi-indices satisfying $\Re S_n \leq A$. Thus the support is locally finite with respect to real asymptotic order. Multiplication, analytic substitution, and coefficient extraction are therefore finite at every prescribed order. This is the elementary finite-grid version of the well-based support condition used in general transseries and Hahn-series calculus.

12 Arbitrary analytic perturbations

The product form is special because its coefficients collapse to a product of binomial factors. The normalization itself needs only analyticity. Let

$$R = e^{\lambda x} \Psi \left(c_1 e^{-\lambda \alpha_1 x}, \dots, c_m e^{-\lambda \alpha_m x} \right), \quad (12.1)$$

where Ψ is analytic near the origin and

$$\Psi(\mathbf{0}) = 1, \quad \Re \alpha_r > 0. \quad (12.2)$$

With $u = e^{-\lambda x}$, $q = R^{-1}$,

$$A_r = c_r u^{\alpha_r}, \quad Q_r = c_r q^{\alpha_r}, \quad V = \frac{u}{q},$$

the inverse equation becomes

$$V = \Psi(\mathbf{A}), \quad A_r = Q_r V^{\alpha_r}, \quad x = \frac{\log R}{\lambda} - \frac{1}{\lambda} \log V. \quad (12.3)$$

Theorem 12.1 (Analytic perturbation inversion). *Under Equation (12.2), there is a unique analytic germ $V(\mathbf{Q}) = 1 + \mathcal{O}(\mathbf{Q})$ solving Equation (12.3). Hence the inverse branch has a convergent transseries*

$$x(R) = \frac{\log R}{\lambda} + \sum_{\mathbf{n} \neq \mathbf{0}} a_n \mathbf{c}^{\mathbf{n}} R^{-S_n}, \quad S_n = \sum_r \alpha_r n_r. \quad (12.4)$$

Its coefficients admit the extraction formula

$$a_{\mathbf{n}} = -\frac{1}{\lambda} \left. \frac{d}{dt} \right|_{t=S_{\mathbf{n}}} [\mathbf{A}^{\mathbf{n}}] \Psi(\mathbf{A})^t \det \left(\delta_{rs} - \alpha_r A_s \frac{\partial}{\partial A_s} \log \Psi(\mathbf{A}) \right). \quad (12.5)$$

Proof. The equations $A_r = Q_r \Psi(\mathbf{A})^{\alpha_r}$ have the solution $\mathbf{A} = \mathbf{0}$ at $\mathbf{Q} = \mathbf{0}$, and their Jacobian with respect to $\mathbf{A}$ is the identity there. The analytic implicit-function theorem gives the unique germ. Apply [Theorem 5.1](#) with $\Phi_r = \Psi^{\alpha_r}$ and auxiliary function Ψ^τ . The coefficient is the expression inside [Equation \(12.5\)](#) evaluated at $t = \tau + S_n$; differentiation at $\tau = 0$, followed by the factor $-1/\lambda$, proves the result. $\square$

The product theorem is recovered by setting $\Psi(\mathbf{A}) = \prod_r (1 - A_r)^{\beta_r}$. Its special rank-one Jacobian and binomial factorization are what turn the general coefficient extraction [Equation \(12.5\)](#) into the closed product [Equation \(11.9\)](#).

12.1 A recursive alternative

For general Ψ , a total-degree recursion may be simpler than [Equation \(12.5\)](#). Write

$$V = 1 + W, \quad W \in (\mathbf{Q}),$$

and iterate

$$W = \Psi(Q_1(1 + W)^{\alpha_1}, \dots, Q_m(1 + W)^{\alpha_m}) - 1 \quad (12.6)$$

by total degree. The right-hand side at degree N depends only linearly on the unknown degree- N part of W , and the coefficient of that part is zero at the origin, so the recursion is triangular. Newton iteration in the complete multi-adic algebra gives quadratic order-doubling when many coefficients are required.

12.2 Uniform analytic estimates

Choose a polydisc $|Q_r| \leq r_r$ on which the implicit solution is analytic and remains away from zeros and branch cuts of Ψ . Cauchy's estimate gives

$$|a_{\mathbf{n}}| \leq M \prod_{r=1}^m r_r^{-n_r} \quad (12.7)$$

for a suitable M . Along a physical ray, $|Q_r| = |c_r| |R|^{-\Re \alpha_r}$ times a bounded angular factor. This immediately supplies uniform truncation estimates by any half-plane of the support, $\Re S_{\mathbf{n}} \leq A$, and justifies termwise differentiation on smaller polydiscs.

Section summary

A dominant exponential plus any analytic combination of finitely many subdominant exponential monomials has a convergent inverse transseries on the additive semigroup generated by their exponents. Product perturbations are the exactly solvable subclass in which multivariate Good inversion collapses to generalized binomial coefficients.

13 Mixed carriers and nested transseries

The preceding theory assumes that the dominant carrier is a pure exponential. Many special functions instead have a carrier of the form

$$C(x) = Ae^{\lambda x} x^\mu (\log x)^\nu \left(1 + \sum_{r \geq 1} \frac{P_r(\log x)}{x^r} \right), \quad (13.1)$$

possibly multiplied by exponentially small factors. Its inverse already belongs to a polynomial-logarithmic or Lambert-type transseries. The log-periodic sectors developed above can be placed over that slower coefficient field.

13.1 Carrier-first normalization

Consider

$$Y = C(x) \Psi(c_1 e^{-\zeta_1 x}, \dots, c_m e^{-\zeta_m x}), \quad \Psi(\mathbf{0}) = 1, \quad \Re \zeta_r > 0. \quad (13.2)$$

Let $x_0 = x_0(Y)$ be a selected inverse branch of the carrier,

$$C(x_0) = Y, \quad (13.3)$$

and write $x = x_0 + \delta$. Define

$$Q_r = c_r e^{-\zeta_r x_0}. \quad (13.4)$$

The exact correction equation is

$$\log C(x_0 + \delta) - \log C(x_0) + \log \Psi(Q_1 e^{-\zeta_1 \delta}, \dots, Q_m e^{-\zeta_m \delta}) = 0. \quad (13.5)$$

If

$$\kappa(x_0) = \left. \frac{d}{dx} \log C(x) \right|_{x=x_0} \quad (13.6)$$

is nonzero, the implicit-function theorem gives a unique correction $\delta = \mathcal{O}(\mathbf{Q})$. Its first term is

$$\delta = -\frac{\log \Psi(\mathbf{Q})}{\kappa(x_0)} + \mathcal{O}(\|\mathbf{Q}\|^2). \quad (13.7)$$

For a product perturbation $\Psi = \prod_r (1 - Q_r)^{\beta_r}$, this becomes

$$\delta = \frac{1}{\kappa(x_0)} \sum_{r=1}^m \beta_r Q_r + \mathcal{O}(\|\mathbf{Q}\|^2). \quad (13.8)$$

The higher coefficients involve derivatives of $\log C$ at x_0 and therefore belong to the differential coefficient field generated by $x_0(Y)$. The small monomials are $e^{-\zeta_r x_0(Y)}$, which may themselves be powers of Y multiplied by powers of $\log Y$ and iterated logarithms. This is the natural nested-transseries construction: invert the slow carrier first, then perform a finite-grid exponential reversion over its coefficient field.

13.2 The carrier $e^{\lambda x} x^\mu$ and Lambert W

Take

$$C(x) = A e^{\lambda x} x^\mu. \quad (13.9)$$

For $\mu \neq 0$, a carrier inverse is

$$x_0(Y) = \frac{\mu}{\lambda} W_k \left(\frac{\lambda}{\mu} \left(\frac{Y}{A} \right)^{1/\mu} \right), \quad (13.10)$$

with the branch W_k selected to match the desired sector. Indeed, raising $e^{\lambda x} x^\mu = Y/A$ to the power $1/\mu$ reduces the equation to the defining form $we^w = z$. When $\mu = 0$,

$$x_0(Y) = \frac{1}{\lambda} \log \frac{Y}{A}. \quad (13.11)$$

The logarithmic derivative is

$$\kappa(x_0) = \lambda + \frac{\mu}{x_0}. \quad (13.12)$$

Consequently the first product correction is

$$\delta = \frac{\sum_r \beta_r c_r e^{-\zeta_r x_0}}{\lambda + \mu/x_0} + \mathcal{O}\left(\max_r |e^{-\zeta_r x_0}|^2\right). \quad (13.13)$$

Expanding W_k at infinity turns the coefficients into polynomials in $\log Y$ and $\log \log Y$, while the factors $e^{-\zeta_r x_0}$ become complex powers of Y decorated by slow logarithmic powers. Thus polynomial-logarithmic and log-periodic transseries are not competing descriptions: the former supplies the slow coefficient field, and the latter supplies the exponentially separated sectors.

13.3 A formal implicit-function principle over a slow field

Let $\mathcal{K}$ be a differential field of slow transseries in a large variable Y , closed under the operations needed to describe $x_0(Y)$ and the derivatives of $\log C$. Let $\mathfrak{m}$ be the ideal generated by the small monomials $Q_r = e^{-\zeta_r x_0}$. If $\kappa(x_0)$ is a unit of $\mathcal{K}$, then Equation (13.5) has a unique solution

$$\delta \in \mathfrak{m} \mathcal{K}[[Q_1, \dots, Q_m]]. \quad (13.14)$$

The proof is the same recursive implicit-function argument as before: the linearized coefficient of δ is the unit $\kappa(x_0)$, and every nonlinear operation raises total Q -degree. This principle is the appropriate abstraction for inverses of Gamma-, Barnes-, hyperfactorial-, and Lambert-type carriers perturbed by subdominant exponential products.

A point requiring care

One should not expand the slow carrier and the exponentially small sectors in a single undifferentiated list of terms. The correct organization is lexicographic: first by exponential sector, then by the internal polynomial-logarithmic scale of its coefficient. Mixing the two orderings can destroy local finiteness and makes remainder statements ambiguous.

14 Complex inverse branches

The real interpolation uses a square root and modulus interpretation, but the squared equation

$$R = e^{\lambda x} + e^{-\lambda x} - 2 \cos(\omega x) \quad (14.1)$$

is entire in x and can be inverted locally in the complex plane. Fix a branch of $\text{Log } R$. The dominant equation $e^{\lambda x} = R$ has logarithmic branches

$$x_{0,m} = \frac{\text{Log } R + 2\pi i m}{\lambda}, \quad m \in \mathbb{Z}. \quad (14.2)$$

Define

$$Z_m = e^{i\omega x_{0,m}}, \quad X_m = R^{-1} Z_m, \quad Y_m = R^{-1} Z_m^{-1}. \quad (14.3)$$

Writing $x = x_{0,m} + \delta$, the normalized equation factors as

$$1 = e^{\lambda \delta} \left(1 - X_m e^{(-\lambda + i\omega)\delta}\right) \left(1 - Y_m e^{(-\lambda - i\omega)\delta}\right). \quad (14.4)$$

With $V = e^{-\lambda \delta}$, this is again

$$A = X_m V^\alpha, \quad B = Y_m V^{\bar{\alpha}}, \quad V = (1 - A)(1 - B).$$

Proposition 14.1 (Right-exponential complex branches). *Whenever $|X_m|$ and $|Y_m|$ are sufficiently small, there is a unique solution near $x_{0,m}$, and it is*

$$x_m(R) = x_{0,m} + \sum_{\substack{j,k \geq 0 \\ j+k \geq 1}} C_{jk} X_m^j Y_m^k, \quad (14.5)$$

with the same universal coefficients C_{jk} as in Equation (5.13).

The condition involves both $|X_m|$ and $|Y_m|$. Since $|Z_m| = \exp(-2\pi\rho m)$ for positive real R and the principal logarithm, moving far in one direction of the branch index makes one generator small but the conjugate generator large. Thus Equation (14.5) describes precisely the complex branches for which the $e^{\lambda x}$ balance remains dominant. Other complex solutions may be governed by a different dominant balance and require a separate Newton polygon or transseries chart.

The real positive branch is $m = 0$, with $Z_0 = e^{i\theta}$, and the real negative branch comes from the exact even symmetry $x \mapsto -x$, not from changing m in the right-exponential family.

15 Interpretation of the log-periodic sectors

15.1 Discrete scale covariance

The phase is

$$\theta = \rho \log R = \pi \log_{\varphi}(\sqrt{5} y). \quad (15.1)$$

A shift $\theta \mapsto \theta + 2\pi$ corresponds to

$$R \mapsto R e^{2\pi/\rho} = R e^{2\lambda} = R \varphi^4, \quad y \mapsto \varphi^2 y. \quad (15.2)$$

Thus every coefficient function $D_n(\theta)$ repeats under the discrete scale factor φ^2 in y , while its radial amplitude acquires the expected factor φ^{-4n} . This is covariance rather than exact periodicity of the whole correction: each sector transforms with its own weight.

Because $D_n(\theta + \pi) = (-1)^n D_n(\theta)$, the half-period scale $y \mapsto \varphi y$ changes the sign of odd sectors and preserves even ones, while multiplying the n -th radial amplitude by φ^{-2n} . This parity mirrors the alternating term in Binet's formula.

15.2 Complex dimensions

The exponents

$$s_{jk} = n - i\rho h \quad (15.3)$$

form a two-dimensional lattice projected onto the complex plane. Their real parts determine decay, and their imaginary parts determine logarithmic frequency. In Mellin-transform language, such exponents play the role of complex poles or complex dimensions. Here they arise algebraically before any transform is taken: they are generated by the two conjugate factors in Equation (3.6). The inverse operation preserves the additive semigroup of these exponents but renormalizes their amplitudes through complex generalized binomial coefficients.

15.3 No Stokes ambiguity on the positive real branch

The expansion is convergent in a nontrivial domain, so its coefficients do not hide an exponentially small ambiguity. On the positive real branch, there is no choice of transseries parameter analogous to a

Stokes constant: analytic continuation from $q = 0$ fixes the solution uniquely. Stokes phenomena may enter only after one enlarges the problem to competing complex dominant balances or to divergent slow-carrier expansions. This distinction is useful: the log-periodic sectors are small relative to inverse logarithms, but they are not nonperturbative ambiguities. They are ordinary convergent Taylor sectors in the lifted variables (X, Y) .

16 Further deductions and research directions

The exact formula opens several directions that are not visible from a few low-order coefficients.

16.1 Pointwise singularity curves

The exact uniform radius q_* is controlled by coefficient growth in the supremum norm over all phases. For a fixed physical phase trajectory $\theta = -\rho \log q$, cancellation among harmonics may postpone the first singularity. Determining the maximal analytic continuation of $q \mapsto \delta(q, -\rho \log q)$ requires solving simultaneously

$$H(\delta; q, \theta) = 0, \quad H_\delta(\delta; q, \theta) = 0, \quad \theta = -\rho \log q \quad (16.1)$$

with compatible logarithm branches. The resulting singularities should govern exponentially accurate large-order asymptotics of the physical one-variable coefficient combinations.

16.2 Asymptotics near the dominant coefficient rays

The maximizers $p_*, 1 - p_*$ in Equation (8.13) identify dominant rays $j/n \rightarrow p_*$ in coefficient space. A two-dimensional saddle-point analysis of $C_{j,n-j}$, followed by a local Fourier summation in j , should produce an asymptotic approximation to $D_n(\theta)$ uniform in θ . Depending on phase, the two conjugate saddles may reinforce or cancel. This would refine the root test into a complete estimate of the prefactor, phase, and transition regions.

16.3 Arithmetic at Fibonacci arguments

Identity Equation (9.3) evaluates the convergent complex-power series at $R = 5F_n^2$ to an integer. The individual terms are generally transcendental-looking, involving $\exp(i\rho h \log(5F_n^2))$, yet their full sum has exact arithmetic meaning. One may seek finite transformations, functional equations, or resummations that make the Binet alternation manifest term by term. The half-period parity Equation (6.9) suggests separating even and odd radial sectors before specializing to Fibonacci arguments.

16.4 Other Lucas sequences

For a Lucas sequence with characteristic roots r and s , a modulus interpolation of $(r^x - s^x)/(r - s)$ usually leads, after squaring, to a dominant exponential plus conjugate or real subdominant exponentials. When the squared expression factorizes into a finite product in $u = e^{-\lambda x}$, Theorem 11.1 applies directly. The nature of the inverse sectors is controlled by the ratios of logarithms of the characteristic roots: real ratios yield generalized powers, while imaginary parts yield log-periodic phases.

16.5 Formalization prospects

The essential proof can be decomposed into formalizable algebraic pieces:

1. the identity $\operatorname{arccsch} 2 = \log \varphi$ and the squared Binet formula;
2. monotonicity of the outer branch on $[2, \infty)$;
3. the factorization [Equation \(3.6\)](#);
4. a formal multivariate power-series implicit-function theorem;
5. Good's coefficient formula and the rank-one determinant calculation;
6. finite coefficient identities in generalized binomial polynomials;
7. analytic convergence and residual bounds on an explicit neighborhood.

The algebraic coefficient theorem is particularly suitable for formalization because every coefficient at a fixed multi-index reduces to a finite polynomial identity; Gamma functions are unnecessary if generalized binomial coefficients are defined by falling factorials.

16.6 Conjectural sharp physical radius

It is natural to distinguish three radii: the bivariate domain of the lifted germ, the uniform Fourier radius q_* , and the radius seen along the physical logarithmic spiral. Numerical analytic continuation could test the following working conjecture.

Conjecture. The first singularity encountered when the positive physical logarithmic-spiral branch is continued inward from arbitrarily small positive q is a square-root branch point produced by the system [Equation \(16.1\)](#); its modulus is at least q_* , with strict inequality unless a maximizing Fourier phase aligns with the logarithmic spiral.

The conjecture is deliberately stated as a program rather than a theorem. The uniform radius proved in [Theorem 8.2](#) is exact in its stated Banach-space sense and does not depend on this pointwise question.

17 Conclusion

The real-argument Fibonacci modulus has an inverse problem whose apparent mixture of an exponential, a reciprocal exponential, and a trigonometric term conceals a simple product structure. After squaring, setting $u = e^{-\lambda x}$, and normalizing by $R = 5y^2$, the entire problem is encoded by

$$Ru = (1 - u^{1-i\rho})(1 - u^{1+i\rho}).$$

The two conjugate factors lift the inverse to a bivariate analytic germ. Multivariate Lagrange–Good inversion then yields the exact coefficient formula [Equation \(5.13\)](#); restriction to the physical logarithmic spiral gives the real Fourier sectors [Equation \(6.6\)](#). The result is simultaneously a complete asymptotic transseries, a convergent expansion, and an effective high-precision algorithm.

Several features are worth retaining as a general template. One should first identify the dominant carrier, then rewrite subdominant terms as a finite set of small exponential monomials, and finally lift those monomials to independent variables before applying formal or analytic inversion. Conjugate complex exponents become periodic functions of the logarithm; rationally independent imaginary parts become quasiperiodic coefficient functions. Product perturbations have closed generalized-binomial coefficients, while arbitrary analytic perturbations retain an explicit Good-determinant extraction formula. When the carrier itself requires a Lambert or polynomial–logarithmic inverse, the same exponential-sector calculus operates over the resulting slow coefficient field.

For the specific interpolation, the two real large-value inverse branches are therefore completely described:

$$X_-(y) = -X_+(y), \quad X_+(y) = \log_\varphi(\sqrt{5}y) + \sum_{n \geq 1} D_n \left(\pi \log_\varphi(\sqrt{5}y) \right) (5y^2)^{-n}.$$

Every coefficient is explicit, every fixed truncation has a certified remainder order, and the full Fourier-sector series has a rigorously identified uniform convergence radius.

A Factorized Fourier coefficients through order six

For convenience, define positive-frequency coefficients by

$$D_n(\theta) = d_{n,0} + 2\Re \sum_{\substack{1 \leq h \leq n \\ h \equiv n \pmod{2}}} d_{n,h} e^{ih\theta}, \quad (\text{A.1})$$

where $d_{n,0}$ is present only for even n . In all formulas below, $\omega = \pi$. The universal expression is

$$d_{n,h} = \frac{(-1)^{n+1}}{\lambda(n - i\rho h)} \binom{n - i\rho h}{(n+h)/2} \binom{n - i\rho h}{(n-h)/2}. \quad (\text{A.2})$$

The following factorizations avoid unnecessary expanded polynomials.

$$\begin{aligned} d_{1,1} &= \frac{1}{\lambda}; \\ d_{2,0} &= -\frac{2}{\lambda}, & d_{2,2} &= -\frac{\lambda - 2i\omega}{2\lambda^2}; \\ d_{3,1} &= \frac{(2\lambda - i\omega)(3\lambda - i\omega)}{2\lambda^3}, & d_{3,3} &= \frac{(\lambda - 3i\omega)(2\lambda - 3i\omega)}{6\lambda^3}; \\ d_{4,0} &= -\frac{9}{\lambda}, \\ d_{4,2} &= -\frac{2(\lambda - i\omega)(2\lambda - i\omega)(3\lambda - 2i\omega)}{3\lambda^4}, \\ d_{4,4} &= -\frac{(\lambda - 4i\omega)(\lambda - 2i\omega)(3\lambda - 4i\omega)}{12\lambda^4}; \\ d_{5,1} &= \frac{(3\lambda - i\omega)(4\lambda - i\omega)^2(5\lambda - i\omega)}{12\lambda^5}, \\ d_{5,3} &= \frac{(\lambda - i\omega)(2\lambda - 3i\omega)(4\lambda - 3i\omega)(5\lambda - 3i\omega)}{8\lambda^5}, \\ d_{5,5} &= \frac{(\lambda - 5i\omega)(2\lambda - 5i\omega)(3\lambda - 5i\omega)(4\lambda - 5i\omega)}{120\lambda^5}; \\ d_{6,0} &= -\frac{200}{3\lambda}, \\ d_{6,2} &= -\frac{(2\lambda - i\omega)(3\lambda - 2i\omega)(3\lambda - i\omega)(5\lambda - 2i\omega)^2}{12\lambda^6}, \\ d_{6,4} &= -\frac{2(\lambda - 2i\omega)(\lambda - i\omega)(3\lambda - 4i\omega)(3\lambda - 2i\omega)(5\lambda - 4i\omega)}{15\lambda^6}, \\ d_{6,6} &= -\frac{(\lambda - 6i\omega)(\lambda - 3i\omega)(\lambda - 2i\omega)(2\lambda - 3i\omega)(5\lambda - 6i\omega)}{60\lambda^6}. \end{aligned}$$

The negative-frequency coefficients are $d_{n,-h} = \overline{d_{n,h}}$. These expressions also provide convenient unit tests for a symbolic implementation of [Equation \(6.6\)](#).

B Direct derivation of the first sectors

This appendix verifies [Equations \(6.10\) to \(6.12\)](#) without multivariate inversion. Substitute

$$\delta = D_1 q + D_2 q^2 + D_3 q^3 + \mathcal{O}(q^4)$$

into [Equation \(3.9\)](#). The coefficient of q is

$$\lambda D_1 - 2 \cos \theta = 0, \quad (\text{B.1})$$

so $D_1 = 2 \cos \theta / \lambda$.

At order q^2 , one obtains

$$\lambda D_2 + \frac{\lambda^2}{2} D_1^2 + 1 + 2\omega D_1 \sin \theta = 0. \quad (\text{B.2})$$

Substituting D_1 and using elementary double-angle identities yields

$$D_2 = -\frac{1}{\lambda} (2 + \cos 2\theta + 2\rho \sin 2\theta).$$

At order q^3 , the coefficient equation is

$$\lambda D_3 + \lambda^2 D_1 D_2 + \frac{\lambda^3}{6} D_1^3 - \lambda D_1 + 2\omega D_2 \sin \theta + \omega^2 D_1^2 \cos \theta = 0. \quad (\text{B.3})$$

After substituting the first two sectors and reducing products of sines and cosines to harmonics 1 and 3, this becomes [Equation \(6.12\)](#). The calculation illustrates both the triangular nature of the scalar recurrence and the rapid growth of algebraic complexity that the closed coefficient formula avoids.

C Residue proof of the Lagrange–Good formula

For completeness, we outline a formal-residue proof of [Theorem 5.1](#). Write

$$Q_r = \frac{A_r}{\Phi_r(\mathbf{A})}. \quad (\text{C.1})$$

The hypothesis $\Phi_r(\mathbf{0}) \neq 0$ makes this a formally invertible change of variables. Coefficient extraction is a multivariate residue:

$$[Q^n]f(\mathbf{A}(\mathbf{Q})) = \text{Res}_{\mathbf{Q}=\mathbf{0}} f(\mathbf{A}(\mathbf{Q})) \prod_{r=1}^m Q_r^{-n_r-1} dQ_r. \quad (\text{C.2})$$

The Jacobian of [Equation \(C.1\)](#) is

$$\det\left(\frac{\partial Q_r}{\partial A_s}\right) = \left(\prod_{r=1}^m \Phi_r^{-1}\right) \det\left(\delta_{rs} - A_r \frac{\partial}{\partial A_s} \log \Phi_r\right). \quad (\text{C.3})$$

By the identity $\det(I - UV) = \det(I - VU)$, the determinant in [Equation \(C.3\)](#) equals

$$\det\left(\delta_{rs} - A_s \frac{\partial}{\partial A_s} \log \Phi_r\right) = \Delta(\mathbf{A}). \quad (\text{C.4})$$

Meanwhile,

$$\prod_r Q_r^{-n_r-1} = \prod_r A_r^{-n_r-1} \Phi_r^{n_r+1}. \quad (\text{C.5})$$

Multiplying [Equation \(C.3\)](#) and [Equation \(C.5\)](#) cancels one factor Φ_r for each r , leaving

$$f(\mathbf{A}) \prod_r \Phi_r^{n_r} \Delta(\mathbf{A}) \prod_r A_r^{-n_r-1} dA_r.$$

Its residue is exactly the coefficient on the right-hand side of [Equation \(5.2\)](#). The same proof is valid analytically on sufficiently small tori or formally in the ring of multivariate Laurent series.

D Notation crosswalk

Symbol	Meaning
φ	Golden ratio $(1 + \sqrt{5})/2$.
λ	Dominant exponential rate $2 \log \varphi$.
ω	Oscillatory frequency π .
ρ	Frequency ratio ω/λ .
R	Squared output variable $5y^2$.
q	Small radial variable R^{-1} .
ξ	Leading inverse carrier $\lambda^{-1} \log R = \log_{\varphi}(\sqrt{5}y)$.
θ	Logarithmic phase $\rho \log R = \pi \xi$.
u	Exponential inverse variable $e^{-\lambda x}$.
$\alpha, \bar{\alpha}$	Conjugate exponents $1 \mp i\rho$.
s_{jk}	Complex radial exponent $j + k - i\rho(j - k)$.
C_{jk}	Bivariate coefficient in Equation (5.13).
$D_n(\theta)$	Real Fourier polynomial multiplying q^n .
S_n	General multi-factor exponent $\sum_r \alpha_r n_r$.

References

- [1] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, “On the Lambert W function,” *Advances in Computational Mathematics* **5** (1996), 329–359.
- [2] P. Flajolet, X. Gourdon, and P. Dumas, “Mellin transforms and asymptotics: harmonic sums,” *Theoretical Computer Science* **144** (1995), 3–58.
- [3] I. M. Gessel, “A combinatorial proof of the multivariable Lagrange inversion formula,” *Journal of Combinatorial Theory, Series A* **45** (1987), 178–195.
- [4] I. M. Gessel, “Lagrange inversion,” *Journal of Combinatorial Theory, Series A* **144** (2016), 212–249; arXiv:1609.05988.
- [5] I. J. Good, “Generalizations to several variables of Lagrange’s expansion, with applications to stochastic processes,” *Proceedings of the Cambridge Philosophical Society* **56** (1960), 367–380.
- [6] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics*, 2nd ed., Addison–Wesley, 1994.
- [7] T. Koshy, *Fibonacci and Lucas Numbers with Applications*, Wiley, 2001.
- [8] F. W. J. Olver, *Asymptotics and Special Functions*, A K Peters, 1997 reprint.
- [9] V. Reshetnikov, “Series and transseries,” in the *ProveIt* Fabius-function research drafts, 2026, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>.
- [10] D. Sornette, “Discrete-scale invariance and complex dimensions,” *Physics Reports* **297** (1998), 239–270.
- [11] J. van der Hoeven, *Transseries and Real Differential Algebra*, Lecture Notes in Mathematics 1888, Springer, 2006.